

AI5707: Pattern Recognition and Machine Learning

Lab1: Familiarising with Matrices, and Distributions

1. Write a program with functions to perform:
 - (a) Matrix multiplication ($m \times n$, $n \times k$) where m , n , k are variable.
 - (b) Dot product of two vectors of dimension n (n is variable).
 - (c) Find the transpose of a matrix of dimension ($m \times n$) (m and n can be variable).
 - (d) Determine whether a given matrix is symmetric.
 - (e) Determine whether a given matrix is upper triangular/lower triangular.
2. Write a program to generate Gaussian random variables.
 - (a) Generate univariate uniform random variables in the range $[a, b]$. Plot the histogram.
 - (b) Use univariate uniform random variables to generate a univariate Gaussian random variable with a given mean (μ , and σ^2).
 - (c) Plot a histogram of the data obtained.

Note: To generate a Gaussian distribution from a uniform random number use the following equations:

$$u_1, u_2 \sim \mathcal{U}(-1, 1), \quad (1)$$

$$s = u_1^2 + u_2^2, \quad (2)$$

where the sample is accepted iff $0 < s < 1$.

Compute the scale factor:

$$k = \sqrt{\frac{-2 \ln s}{s}}. \quad (3)$$

Finally generate two independent Gaussian Random variables:

$$x = u_1 k = u_1 \sqrt{\frac{-2 \ln s}{s}}, \quad (4)$$

$$y = u_2 k = u_2 \sqrt{\frac{-2 \ln s}{s}}. \quad (5)$$

x and y are both Gaussian random variables with 0 mean and variance 1.

To generate a Gaussian random variable with $\mathcal{N}(\mu, \sigma^2)$ use the following:

$$x' = \mu + \sigma x, \quad (6)$$

$$y' = \mu + \sigma y. \quad (7)$$